GIS PRACTICAL REPORT | CA4 | QGIS + GOOGLE EARTH ENGINE

Urban Climate Stress Mapping of Navi Mumbai

One ward-level index that combines heat, lack of greenery and flood susceptibility for 111 wards

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Subject / faculty	Geographic Information Systems (GIS), faculty: Vijaya Ahire
Software	QGIS 4.2 and Google Earth Engine Code Editor (JavaScript)
Project files	NaviMumbai_UCSI.qgz (project), NaviMumbai_UCSI.gpkg (spatial database), maps, query results, this report
Data period	Satellite images of April 2026. Written September 2026.

Summary in one minute

What: a map that ranks the 111 wards of Navi Mumbai by combined stress from heat, few plants and flood risk (the Urban Climate Stress Index, UCSI).

Result: 23 wards fall in the highest class. They cover only 8.1 km² (6.6% of the city) but are the hottest (45.2 °C ward average), most built-up (82%) and least green (NDVI 0.18). Top five wards: 70, 57, 67, 11 and 29.

Database and queries: one GeoPackage with 7 spatial layers, 1 table and 2 QGIS relations; 5 attribute queries, 5 spatial queries and 1 overlay analysis, all saved as layers.

Honest limits: one season (April 2026), a susceptibility index (not a flood forecast) and the Earth Engine scripts have not been run yet (see Section 14).

1. Aim

To build a basic GIS application in QGIS that maps urban climate stress for the wards of Navi Mumbai. The application stores the data in a GeoPackage, answers attribute and spatial queries, and ends in a final map that shows which wards need attention first.

2. Objectives

- Model spatial data with polygon, line and point layers and suitable attribute fields (Part A of the brief).
- Create a spatial database in GeoPackage with Integer, Text, Decimal and Date fields, and define relations between tables (Part B).
- Measure three stresses for every ward from free satellite data: heat, lack of greenery and flood susceptibility.
- Combine the three stresses into one index (UCSI), and test whether the ranking stays stable when the weights change.
- Run 5 attribute queries and 5 spatial queries, plus one overlay analysis (Parts C and D).

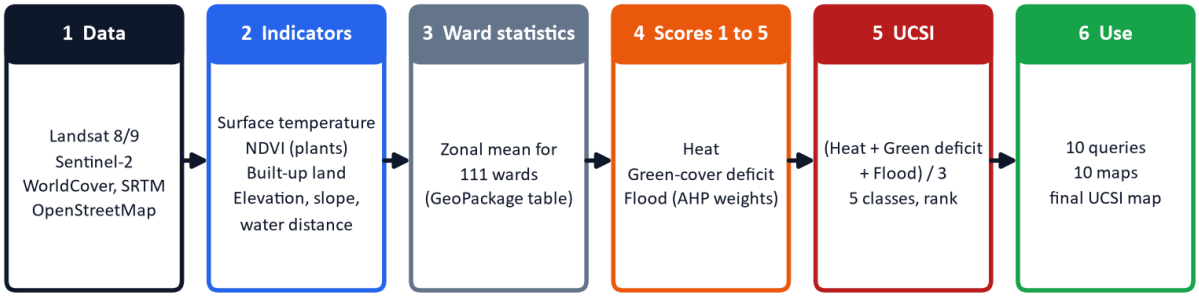
- Prepare thematic maps and a final map layout with title, legend, north arrow, scale bar, labels and sources (Part E).

3. Problem statement

Navi Mumbai is a planned city on the west coast of India, between Thane Creek and the Parsik hills. Three problems affect its people at the same time: hot surfaces, few plants and trees, and low ground that collects water in heavy rain. These are usually mapped one by one, so it is hard for a ward office to say which wards need attention first.

This project builds one simple index for each of the **111 wards**: **UCSI = (Heat score + Green-cover deficit score + Flood susceptibility score) ÷ 3**. Every score runs from 1 (least stress) to 5 (most stress), so the three parts can be added fairly.

A real event gives the topic a practical reason. On 26 May 2025 unusual pre-monsoon rain fell on Navi Mumbai. News reports gave 155.7 mm at Nerul and 133.9 mm at Belapur in one day, and the APMC market at Vashi was reported waterlogged (Free Press Journal, 26 May 2025). This report uses the event only as a rough check of the flood layer. It is not used to calibrate the model.



Google Earth Engine scripts (folder gee/) collect steps 1 to 3. QGIS builds steps 4 to 6.

Figure 1. Workflow of the project, from open satellite data to the final UCSI map.

4. Software and data used

Software

Software	Used for
QGIS 4.2	Project, GeoPackage, symbology, labels, Field Calculator, Select by Expression / Location, Processing tools, print layouts.
Google Earth Engine (Code Editor)	Five JavaScript scripts in the folder gee/ that collect LST, NDVI, built-up land, terrain and ward statistics from public data catalogues.
Python 3 (rasterio, geopandas, numpy)	Used to process the same public datasets into the delivered rasters and ward statistics, outside Earth Engine (see Section 14).
OpenStreetMap Overpass API	Roads and hospitals inside the study area.

Data

All data are free and open. No paid data or software was used.

Data	Source	Details	Used for
Ward boundaries	OpenCity / BharatLAS public ward files	111 ward polygons. Two public versions were compared and one was used (wards_32643). Not an official NMMC file.	Unit of analysis
Surface temperature (LST)	Landsat 8 and 9, Collection 2 Level 2, band ST_B10	3 clear scenes: 13 Apr (L8), 22 Apr (L8), 30 Apr 2026 (L9). 30 m grid; the thermal sensor itself sees about 100 m.	Heat score
Vegetation (NDVI)	Sentinel-2 L2A, bands B4 and B8, cloud mask SCL	2 scenes: 8 and 11 April 2026. 10 m.	Green-cover deficit
Land cover	ESA WorldCover 2021 v200	10 m classes: built-up, tree, water, mangrove, etc.	Built-up %, water bodies, distance to water
Elevation	SRTM 1 arc-second (NASA / USGS)	About 30 m. Slope calculated from it.	Flood score
Roads and hospitals	OpenStreetMap contributors	93 hospitals and clinics inside the study area; 729 major road segments (motorway, trunk, primary, secondary).	Queries S2 to S4
Flood reports	News report (Free Press Journal, 26 May 2025)	3 points digitised by hand: Nerul and Belapur rainfall, APMC market waterlogging. Positions are approximate.	Check of flood layer

5. Spatial data model

The project uses **WGS 84 / UTM zone 43N (EPSG:32643)**. Navi Mumbai lies in UTM zone 43N and the unit is the metre, so areas in hectares and buffers of 100 or 200 m are correct.

Layer	Geometry	Count	Key attributes	How it was created
study_boundary	Polygon	1	Boundary_ID, Name, N_Wards, Area_km2, Source, Created (Date)	Dissolved from the 111 wards
wards	Polygon	111	Ward_ID (key), Ward_Name, LGD_Code, Area_km2, LST_C, NDVI, BuiltUp_Pct, Elev_m, Slope_Pct, DistWater_m, F_Elev, F_Slope, F_Water, F_BuiltUp, Heat_Score, GreenDef_Score, Flood_Score, UCSI, UCSI_Class, UCSI_Rank, Period_Start, Period_End	Imported, then zonal statistics and formulas
water_bodies	Polygon	100	Water_ID, Water_Type (Water / Mangrove / Wetland), Area_ha, Source	Vectorised from WorldCover (0.5 ha or more)
major_roads	Line	729	Road_ID, Road_Name, Road_Type, Ref, Length_km, Flooded_2025, Source	OpenStreetMap
hospitals	Point	93	Hosp_ID, Name, Type, Ward_ID, Source	OpenStreetMap, ward found by spatial join
flood_reports	Point	3	Report_ID, Site, Report_Type, Event_Date (Date), Rain_mm, Note, Source	Digitised from a news report
hot_lowgreen_zones	Polygon	71	Zone_ID, Area_ha, Mean_LST_C, Mean_NDVI, Rule	Raster overlay: LST 45 °C or more AND NDVI below 0.2; patches of 2 ha or more
ward_lst_by_date	Table	333	Ward_ID, Scene_ID, Scene_Date (Date), Platform, LST_Mean_C, Valid_Pixels	Ward mean LST for each of 3 dates

The GeoPackage also holds nine GeoTIFF rasters (data/rasters): *LST_Apr2026_C, NDVI_Apr2026, LandCover_WorldCover2021, BuiltUp_WorldCover2021, Elevation_SRTM_m, Slope_SRTM_pct, Distance_to_Water_m, HeatScore_1to5, GreenDeficit_1to5*.

Honest note on digitising: the ward polygons and the OpenStreetMap features were imported, not drawn by hand. The only points digitised by hand are the three flood reports.

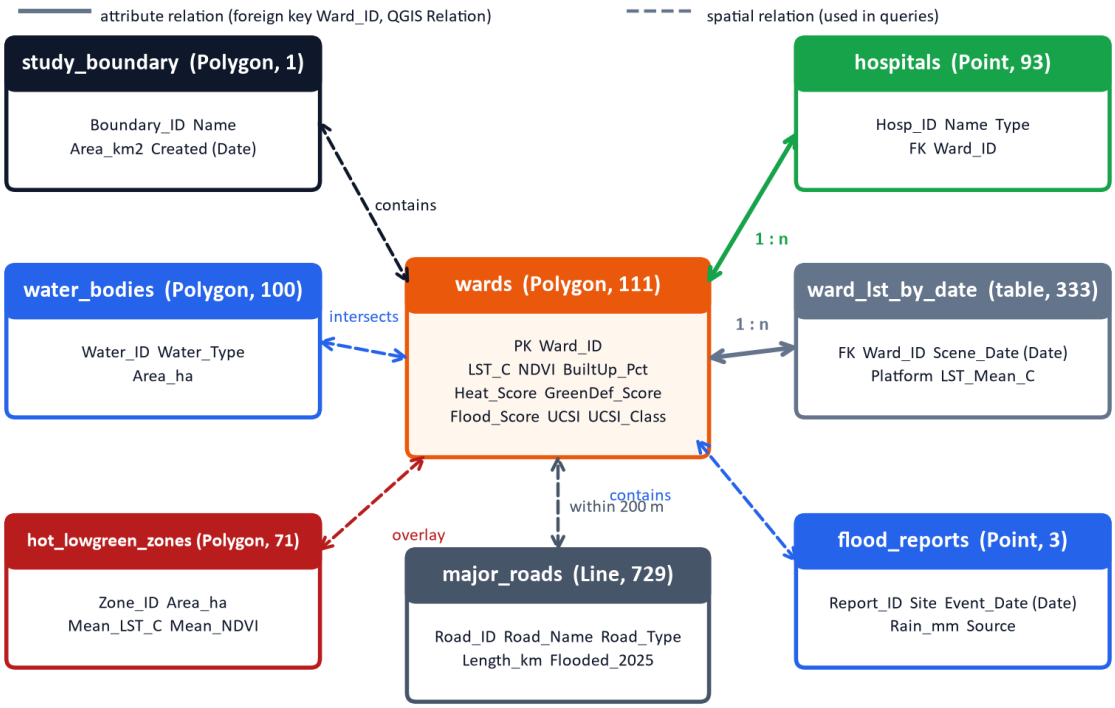


Figure 2. Spatial data model. Solid lines are attribute relations (key Ward_ID). Dashed lines are spatial relations used in the queries.

6. Database design

A single GeoPackage, **NaviMumbai_UCSI.gpkg**, stores all vector layers and the table. A second one, **NaviMumbai_query_results.gpkg**, stores the output of every query so that results can be reopened in QGIS.

Field type	Example fields	Why this type
Integer	Ward_ID, Hosp_ID, F_Elev, UCSI_Rank	Whole numbers, keys and counts.
Text	Ward_Name, Water_Type, UCSI_Class, Flooded_2025	Names and categories.
Decimal	LST_C, NDVI, BuiltUp_Pct, UCSI, Rain_mm	Measured values and scores. Stored with 1 to 4 decimals so that formulas can be repeated in QGIS.
Date	Period_Start, Period_End, Scene_Date, Event_Date, Created	Dates of images and events.

Relations. Ward_ID links the tables. Two relations are defined in QGIS (Project > Properties > Relations): **wards to ward_lst_by_date** (one ward, three dates) and **wards to hospitals** (one ward, many hospitals). In the attribute form of a ward, QGIS shows its child rows. A spatial index is created on every layer.

Integrity checks. All 93 hospitals have a valid Ward_ID (0 unmatched). The table ward_lst_by_date has 333 rows, which is exactly 111 wards × 3 dates. Invalid polygon geometries created during vectorising (24 water polygons, 34 hot-zone polygons) were repaired with MakeValid before the queries were run.

7. Procedure

Each step names the QGIS menu or the Earth Engine script. Steps 3 to 5 follow the scripts in the folder gee/.

1. **Create the project and set the CRS.** Project > New, Project > Properties > CRS = EPSG:32643. Save as NaviMumbai_UCSI.qgz with relative paths (Project > Properties > General).
2. **Load and check the wards.** Add the 111 ward polygons. Run Vector > Geometry Tools > Check Validity. Keep the key as Ward_ID.
3. **Land surface temperature (script 01).** Landsat 8 and 9 Level 2, 10 April to 1 May 2026, cloud cover below 20%. Remove cloud, cirrus and shadow pixels with QA_PIXEL bits 1 to 4. Convert the band: LST (°C) = ST_B10 × 0.00341802 + 149.0 − 273.15. Take the median of the clear scenes.
4. **NDVI (script 02).** Sentinel-2 L2A, 8 to 13 April 2026. Remove cloud, shadow and cirrus with the SCL band (classes 3, 8, 9, 10). NDVI = (B8 − B4) ÷ (B8 + B4). Take the median.
5. **Built-up land, terrain and water distance (scripts 03 and 04).** WorldCover class 50 is built-up. Slope in % comes from SRTM. Distance to water and mangrove (classes 80 and 95) is measured in metres.
6. **Ward statistics (script 05, or Raster > Zonal statistics in QGIS).** Mean value of every indicator for each ward, joined to the wards layer on Ward_ID.
7. **Scores and index (Field Calculator).** Formulas below. Heat and green-cover scores are stretched to 1 to 5 across the 111 wards. The flood score uses AHP weights.
8. **Other layers.** Water and mangrove polygons, hot and low-green patches (Raster calculator, then Raster > Conversion > Polygonize), roads and hospitals from OpenStreetMap, three digitised flood-report points. Write all layers to the GeoPackage with declared field types.
9. **Symbology and labels.** Graduated (Natural Breaks) for the score maps, Categorized for UCSI classes, Rule-based for roads, Pseudo-colour for rasters. Ward number labels with a white halo.
10. **Queries.** Select by Expression, Select by Location, Buffer, Centroids, Join attributes by nearest and Intersection (Sections 8 to 10). Results are saved to NaviMumbai_query_results.gpkg.
11. **Layouts.** Project > New Print Layout (A4). Map frame, legend, north arrow, scale bar, title, source line and a key-numbers box. Exported as PNG.

Formulas used in the Field Calculator

```
Heat_Score      = 1 + 4 * ("LST_C" - minimum("LST_C")) / (maximum("LST_C") - minimum("LST_C"))
GreenDef_Score  = 5 - 4 * ("NDVI" - minimum("NDVI")) / (maximum("NDVI") - minimum("NDVI"))
Flood_Score     = 0.42486*"F_Elev" + 0.16129*"F_Slope" + 0.27007*"F_Water" + 0.14378*"F_BuiltUp"
UCSI            = ("Heat_Score" + "GreenDef_Score" + "Flood_Score") / 3
UCSI_Class      = CASE WHEN "UCSI_Rank" <= 23 THEN 'Highest' WHEN "UCSI_Rank" <= 45 THEN 'High'
                  WHEN "UCSI_Rank" <= 67 THEN 'Medium' WHEN "UCSI_Rank" <= 89 THEN 'Low' ELSE
'Lowest' END
```

UCSI_Rank is the position of the ward when all 111 wards are sorted by UCSI from highest to lowest (1 = highest stress). The five classes hold 23, 22, 22, 22 and 22 wards (quantiles). These formulas were tested in QGIS against the stored values and agree to within 0.01 (rounding of stored fields).

Flood susceptibility: factor classes and AHP weights

Factor	Class limits and score (5 = most susceptible)	Weight	Field
Elevation (m)	≤ 5 → 5; ≤ 10 → 4; ≤ 20 → 3; ≤ 50 → 2; > 50 → 1	0.425	F_Elev
Slope (%)	≤ 2 → 5; ≤ 5 → 4; ≤ 10 → 3; ≤ 20 → 2; > 20 → 1	0.161	F_Slope
Distance to water / mangrove (m)	≤ 250 → 5; ≤ 500 → 4; ≤ 1000 → 3; ≤ 2000 → 2; > 2000 → 1	0.270	F_Water
Built-up land (%)	≥ 60 → 5; ≥ 40 → 4; ≥ 20 → 3; ≥ 10 → 2; < 10 → 1	0.144	F_BuiltUp

The weights come from an AHP pairwise comparison (Saaty scale) in the order elevation, slope, water distance, built-up: rows `[1, 2, 2, 3]` `[1/2, 1, 1/2, 1]` `[1/2, 2, 1, 2]` `[1/3, 1, 1/2, 1]`. The consistency ratio is **0.017** (below 0.10 means the judgements are consistent). The class limits and the pairwise judgements are the author's choices based on common practice. They are assumptions, not measured facts, which is why Section 10 tests them.

8. Attribute queries

Run with Select by Expression (or Processing > Extract by expression). Each result is a layer in NaviMumbai_query_results.gpkg and a page in Appendix D.

No.	Question	QGIS expression	Wards	Highlight
A1	Wards that are 80% or more built-up	`"BuiltUp_Pct" >= 80`	**30**	Ward 70 is the most built-up (98.3%).
A2	Wards with very few plants	`"NDVI" < 0.2`	**24**	Ward 70 has the lowest NDVI (0.116).
A3	Wards with average surface temperature of 45 °C or more	`"LST_C" >= 45`	**23**	Ward 70 is the hottest (47.6 °C).
A4	Wards with flood susceptibility score of 4.5 or more	`"Flood_Score" >= 4.5`	**22**	Wards 16, 33 and 77 score highest (4.84).
A5	Wards that are very built-up AND very bare	`"BuiltUp_Pct" >= 80 AND "NDVI" < 0.2`	**18**	Ward 70 ranks first, then ward 57 (UCSI rank 2).

Reading the results. A1 to A3 find the same core of wards again and again, which is expected: more built-up land means fewer plants and hotter ground. Correlation across the 111 wards is r = +0.76 between built-up share and temperature, and r = -0.75 between NDVI and temperature (Figure 3). A5 shows the overlap directly: 18 wards are both very built-up and very bare.

Wards in A5: 11, 28, 38, 40, 43, 44, 47, 50, 57, 66, 67, 70, 75, 81, 86, 89, 93, 95.

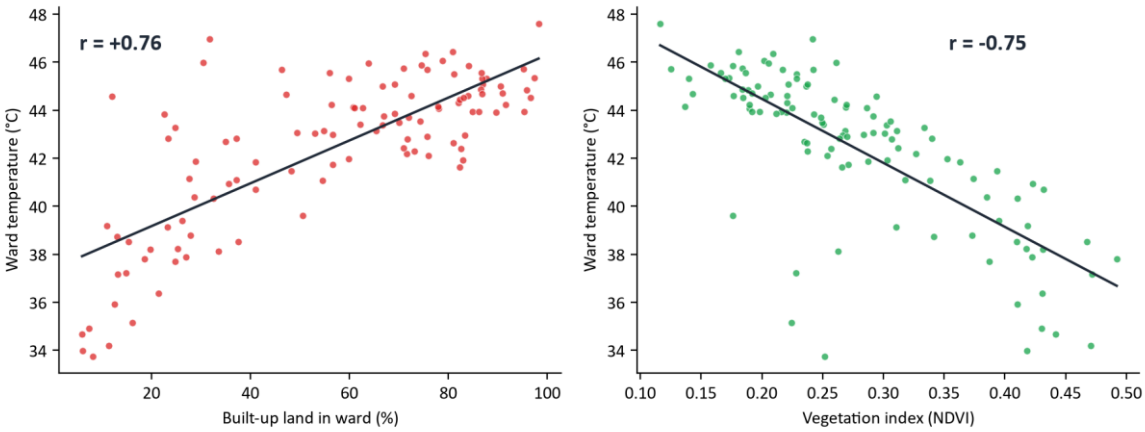


Figure 3. Each dot is one ward. More built-up land and less vegetation go with hotter surfaces.

9. Spatial queries

Run with Vector > Research Tools > Select by Location and the Geoprocessing tools. The word in brackets is the spatial predicate.

No.	Question (predicate)	QGIS method	Result
S1	Wards that touch water or mangrove of 1 ha or more (intersect)	Select by Location: wards intersect water_bodies where Area_ha >= 1	48 of 111 wards
S2	Wards within 200 m of a major road (buffer + intersect)	Buffer (200 m, dissolve) on motorway, trunk and primary roads, then Select by Location: wards intersect the buffer	78 of 111 wards
S3	Hospitals inside the highest-stress wards (within)	Extract the wards where UCSI_Class = Highest, then Select by Location: hospitals are within those wards	8 of 93 hospitals, in 23 wards
S4	Nearest hospital to each highest-stress ward (nearest feature)	Centroids, then Join attributes by nearest (1 neighbour); distance in metres	mean 693 m, shortest 74 m, longest 2505 m
S5	Ward that contains the waterlogged APMC site (contains)	Select by Location: wards contain the flood_reports point where Report_Type = Waterlogged site	ward 57

What the spatial queries show

- **S1:** 48 of 111 wards touch water or mangrove. Distance to water is one of the four flood factors, so these wards are checked first for flood exposure.
- **S2:** 78 wards have a motorway, trunk or primary road within 200 m. Two roads in the layer, Thane–Belapur Road and Sion–Panvel Highway, were reported waterlogged on 26 May 2025 (66 segments are flagged in the field Flooded_2025).
- **S3:** Only 8 of the 93 mapped hospitals and clinics are inside the 23 highest-stress wards: Mody Hospital Iccu (ward 12); Sai Seva Hospital and Icu, Mumbai (ward 13); Indravati Hospital (ward 13); Sanjivni Hospital, Thane (ward 13); Unnamed hospital (ward 13); NMMC MCH Airoli (ward 13); Aakashdeep Hospital and I.C.U (ward 44); Healing Hands Clinic - Vashi, New Mumbai (ward 57). OpenStreetMap may be missing some facilities, so this is a first check, not a full health-access study.
- **S4:** From the centre of each highest-stress ward the nearest mapped hospital is 693 m away on average. The longest distance is 2505 m (ward 2). Distances are straight lines, not road distances.

- **S5:** The waterlogged APMC market site lies in ward 57. The model gives this ward a flood score of 4.57, a UCSI of 4.48 and rank 2 of 111. The news report and the model agree here. This is a small consistency check on one point, not proof.

10. Spatial analysis

A. Summary by UCSI class

Class	Wards	Area (km²)	Area (%)	Mean LST (°C)	NDVI	Built-up (%)	Flood	UCSI range
Highest	23	8.1	6.6	45.2	0.18	82	4.18	4.13 to 4.48
High	22	9.6	7.8	44.4	0.22	77	4.10	3.87 to 4.12
Medium	22	11.2	9.1	43.1	0.27	63	4.08	3.53 to 3.86
Low	22	46.9	38.1	41.3	0.31	38	3.68	3.06 to 3.51
Lowest	22	47.1	38.3	37.9	0.41	21	3.97	2.25 to 3.04

The wards with the highest stress are small and dense: 23 wards cover 8.1 km², while the two lowest classes cover 94.0 km² (76.4% of the city). Ward-average temperature rises from 37.9 °C in the lowest class to 45.2 °C in the highest, and NDVI falls from 0.41 to 0.18.

B. Top 10 wards

Rank	Ward	UCSI	Heat	Green deficit	Flood	LST (°C)	NDVI	Built-up (%)	Area (ha)
1	**70**	**4.48**	5.00	5.00	3.45	47.6	0.12	98	7
2	**57**	**4.48**	4.49	4.36	4.57	45.8	0.18	84	156
3	**67**	**4.43**	4.67	4.31	4.30	46.4	0.18	81	118
4	**11**	**4.41**	4.34	4.75	4.14	45.3	0.14	88	17
5	**29**	**4.40**	4.01	4.78	4.41	44.2	0.14	78	23
6	**66**	**4.39**	4.34	4.43	4.41	45.3	0.17	87	18
7	**89**	**4.31**	4.46	4.91	3.56	45.7	0.13	95	13
8	**28**	**4.30**	4.13	4.36	4.41	44.6	0.18	84	22
9	**3**	**4.29**	4.15	4.03	4.69	44.7	0.21	47	123
10	**16**	**4.27**	4.27	3.71	4.84	45.1	0.24	69	20

Ward 70 is very small (about 7 ha, roughly 8 thermal pixels), so its top rank is less certain than that of larger wards such as 57 or 67. Wards 57 and 66 are in the top quarter for all three stresses at the same time.

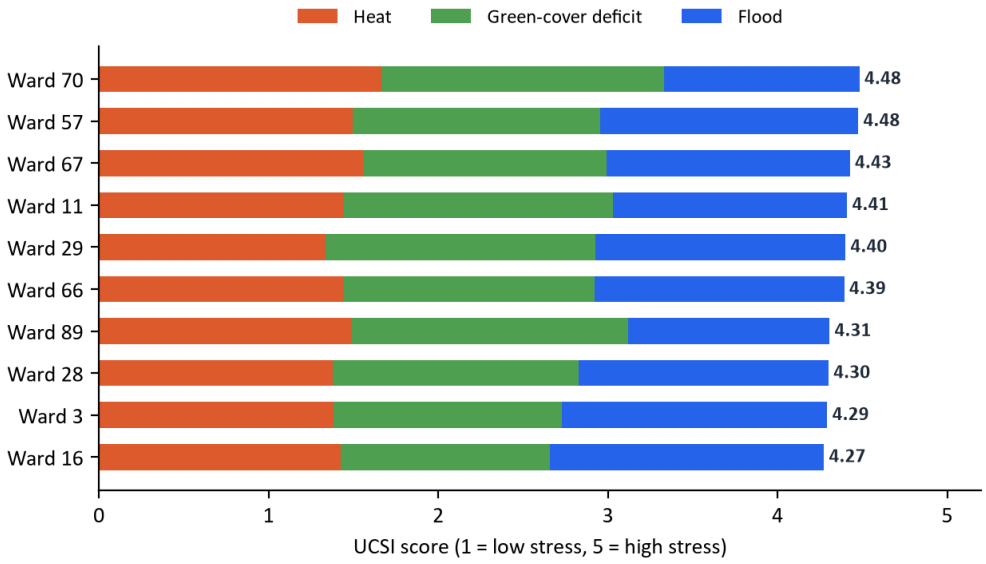


Figure 4. What makes up the UCSI of the ten highest-ranked wards.

C. Overlay analysis: hot AND low-green land (S6)

A raster overlay of LST 45 °C or more AND NDVI below 0.2 gives 18.9 km² of pixels (15.4% of the study area). Patches of 2 ha or more (71 patches, 17.6 km²) were turned into the layer hot_lowgreen_zones and intersected with the wards. 85 of 111 wards contain some of it. Ward 26 has the largest area (282 ha, 16.7% of the ward). Among wards larger than 50 ha, ward 67 has the highest share (60.3% of the ward). These patches are the first places to look at for tree planting or cool roofs.

D. How stable is the ranking?

UCSI gives equal weight to the three parts. Two alternatives were tested: heat weighted 50% (heat 0.50, green 0.25, flood 0.25) and flood weighted 50%. The rank correlation (Spearman) with the equal-weight ranking is 0.983 and 0.95. Seven of the ten highest wards stay in the top ten when heat is weighted more, and eight of ten when flood is weighted more. The ranking is therefore stable, but the exact order inside the top ten can change.

11. Results

- **23 wards form the highest-stress class.** They cover only 8.1 km² but have the hottest ground (45.2 °C ward average), the most built-up land (82%) and the least vegetation (NDVI 0.18).
- **Top five wards: 70, 57, 67, 11 and 29** (UCSI 4.48, 4.48, 4.43, 4.41, 4.40).
- **Built-up land explains much of the heat.** r = +0.76 (built-up share) and r = -0.75 (NDVI) against ward temperature.
- **Ward 57 stands out.** It contains the APMC market that was waterlogged on 26 May 2025, has a flood score of 4.57, and is in the top quarter for heat, green-cover deficit and flood at the same time.
- **Cool wards** are on the western creek and mangrove side and in the eastern hills. The lowest UCSI is ward 96 (2.25).
- **Hospital access is thin where stress is high.** 8 of 93 mapped hospitals are in the 23 highest wards, and the nearest hospital is 693 m from a ward centre on average.
- **The ranking is stable** when the weights change (rank correlation 0.98 and 0.95).

12. Final map

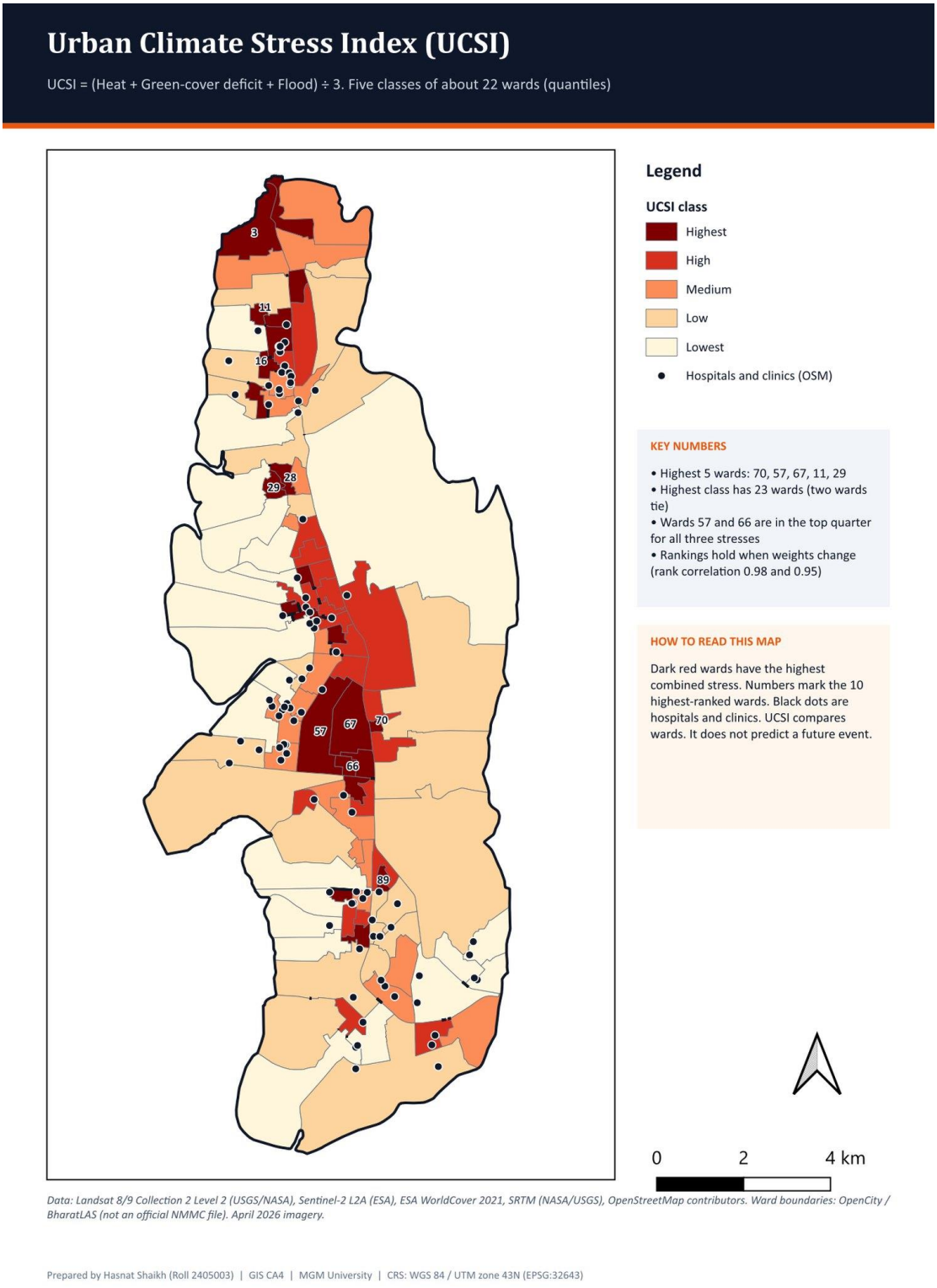


Figure 5. Final map: Urban Climate Stress Index (UCSI) of the 111 wards of Navi Mumbai. Numbers mark the ten highest-ranked wards. Layout exported from QGIS.

13. GIS application and project files

The QGIS project is the application. A user works in four moves, as the brief asks: Select, Query, Analyse, Visualize.

Move	What the user does in NaviMumbai_UCSI.qgz
Select	Click a ward with the Identify tool, or use Select by Expression on the layer wards. The layer tree groups results, analysis layers, boundaries, rasters, tables and saved query results.
Query	Open Select by Expression and type any of the expressions in Section 8, or use Select by Location for the queries in Section 9. The relation panel of a ward lists its three LST dates and its hospitals.
Analyse	Processing Toolbox: Buffer, Intersection, Join attributes by nearest, Zonal statistics. Field Calculator to change weights and recompute UCSI.
Visualize	Ten map themes (Map Themes button), four bookmarks, and 21 saved print layouts (10 maps and 11 query pages) in Project > Layouts.

Files delivered

File or folder	Content
`NaviMumbai_UCSI.qgz`	QGIS project (relative paths, EPSG:32643, 2 relations, 10 themes, 4 bookmarks, 21 layouts)
`data/NaviMumbai_UCSI.gpkg`	Spatial database: 7 spatial layers + 1 table (typed fields)
`data/NaviMumbai_query_results.gpkg`	Saved output of all queries and helper layers (14 layers)
`data/rasters/`	Nine GeoTIFF rasters (LST, NDVI, land cover, built-up, elevation, slope, water distance, heat and green scores)
`maps/`	Ten map layouts as PNG (Map1 to Map10; Map9 is the final map)
`queries/`	Eleven query result pages (A1 to A5, S1 to S6), Query_summary.csv, query_results.json
`gee/`	Five Google Earth Engine scripts and README.txt (how to run and what values to expect)
`report/`	This report (Word and PDF)

14. Interpretation and limits

A good result is one that says clearly what it can and cannot show. These are the limits of this project.

- **Google Earth Engine.** The five scripts were written from the official dataset pages (dataset names, bands and scale factors checked) but they have not been run yet. The delivered rasters were made from the same public datasets and the same dates, using Python outside Earth Engine. When the scripts are run, the results should agree within about 1 to 2 °C for LST and 0.03 for NDVI (values to check are listed in gee/README.txt).
- **One season.** All images are from April 2026, before the monsoon and near the hottest time of the year. Greenery is lower and heat higher than in the monsoon. The results describe the pre-monsoon pattern only.
- **Surface temperature is not air temperature.** The satellite sees roofs, roads and soil. Dry bare hills in the east can look hot even though nobody lives there, so heat alone can mislead in such areas.
- **Thermal resolution.** The thermal sensor sees about 100 m and the product is resampled to 30 m. Very small wards, such as ward 70 (about 7 ha), contain only a few thermal pixels, so their scores are less reliable.

- **The flood score is a susceptibility index, not a flood forecast.** It uses terrain and land cover only. SRTM includes buildings and trees. There is no drainage, tide, soil or rainfall data (rainfall from a 5 km grid would give only about four pixels for the whole city). Class limits and AHP judgements are the author's assumptions, checked with the consistency ratio (0.017) and the weight test in Section 10.
- **One local check only.** The waterlogged APMC site falls in a ward with a high flood score, but the Nerul rain point falls in a ward with a moderate score. Three news points cannot validate a model.
- **UCSI is relative.** Scores are stretched across the 111 wards. A score of 5 means "worst in Navi Mumbai", not an absolute danger level.
- **Boundaries and areas.** The ward polygons come from public files, not from an official NMMC dataset, and public sources quote different city areas. This report uses 122.9 km², the sum of the 111 ward polygons. The eight NMMC ward offices are not in the data, so results are not grouped by ward office. This can be added later as a Text field.
- **OpenStreetMap is volunteer data.** Some hospitals and roads may be missing, and distances are straight lines.

15. Conclusion

A basic GIS application was built in QGIS for the 111 wards of Navi Mumbai. It has a spatial database in GeoPackage with typed fields and relations, ten queries and one overlay analysis, ten thematic maps and a final map. The Urban Climate Stress Index shows that stress is concentrated in a small number of dense central wards: 23 wards cover 8.1 km² but are the hottest, the most built-up and the least green, and wards 70, 57, 67, 11 and 29 rank highest. Ward rankings stay stable when the weights change, and the overlay of hot and low-green land gives a short list of places for tree planting and cool roofs. The results are a first screening based on one season and open data. The next steps are to run the Earth Engine scripts, add a monsoon-season image, and use an official ward-office list.

16. References and data sources

- USGS. Landsat Collection 2 Level 2 Science Products (surface temperature, band ST_B10). usgs.gov/landsat-missions
- ESA / Copernicus. Sentinel-2 Level-2A surface reflectance. sentinels.copernicus.eu
- Zanaga, D. et al. (2022). ESA WorldCover 10 m 2021 v200. esa-worldcover.org
- NASA / USGS. SRTM 1 arc-second Global elevation. Earth Engine dataset USGS/SRTMGL1_003
- OpenStreetMap contributors. Road and hospital data via the Overpass API. openstreetmap.org
- OpenCity and BharatLAS. Navi Mumbai ward boundary files (public).
- Free Press Journal (26 May 2025). News report on heavy pre-monsoon rain and waterlogging in Navi Mumbai.
- Saaty, T. L. (1980). The Analytic Hierarchy Process. McGraw-Hill.
- Google Earth Engine dataset catalogue. developers.google.com/earth-engine/datasets
- QGIS Project. QGIS User Guide. docs.qgis.org
- Data accessed through public catalogues: Microsoft Planetary Computer (Landsat), AWS Earth Search (Sentinel-2) and AWS Open Data (WorldCover, SRTM).

Appendix A. Ward table (all 111 wards)

Values as stored in the layer wards. LST and NDVI are ward means. Scores run from 1 (least stress) to 5 (most stress). Rank 1 = highest stress.

Ward	LST (°C)	NDVI	Built-up (%)	Heat	Green def.	Flood	UCSI	Rank	Class
1	47.0	0.242	31.7	4.82	3.66	2.10	3.53	67	Medium
2	45.6	0.166	55.9	4.41	4.47	3.58	4.15	21	Highest
3	44.7	0.207	47.2	4.15	4.03	4.69	4.29	9	Highest
4	45.7	0.242	46.3	4.45	3.66	2.83	3.65	62	Medium
5	46.0	0.261	30.4	4.53	3.46	2.10	3.36	74	Low
6	44.6	0.294	12.1	4.13	3.11	1.95	3.06	89	Low
7	45.7	0.217	75.8	4.45	3.93	3.72	4.03	33	High
8	45.7	0.184	71.0	4.46	4.28	3.99	4.24	12	Highest
9	41.9	0.287	28.9	3.34	3.18	4.55	3.69	59	Medium
10	41.8	0.363	41.1	3.34	2.37	4.27	3.33	76	Low
11	45.3	0.140	87.7	4.34	4.75	4.14	4.41	4	Highest
12	45.1	0.222	86.9	4.27	3.88	4.41	4.19	17	Highest
13	45.0	0.236	66.7	4.25	3.72	4.57	4.18	18	Highest
14	37.8	0.492	18.6	2.18	1.00	4.41	2.53	105	Lowest
15	39.4	0.395	26.2	2.63	2.03	4.55	3.07	88	Low
16	45.1	0.237	69.2	4.27	3.71	4.84	4.27	10	Highest
17	45.5	0.229	81.2	4.39	3.81	4.14	4.11	26	High
18	43.8	0.291	66.9	3.89	3.14	3.98	3.67	60	Medium
19	45.3	0.229	59.8	4.34	3.80	3.10	3.75	54	Medium
20	44.1	0.269	60.8	4.00	3.37	2.70	3.36	75	Low
21	43.4	0.251	62.2	3.79	3.57	3.98	3.78	51	Medium
22	44.1	0.190	78.0	3.98	4.22	4.41	4.20	15	Highest
23	41.1	0.318	37.1	3.13	2.85	4.55	3.51	68	Low
24	39.2	0.419	11.0	2.57	1.78	4.41	2.92	95	Lowest
25	41.2	0.374	27.6	3.14	2.26	4.13	3.17	84	Low
26	43.3	0.334	24.8	3.75	2.69	1.99	2.81	98	Lowest
27	44.1	0.286	61.1	3.99	3.19	4.41	3.86	46	Medium
28	44.6	0.177	84.0	4.13	4.36	4.41	4.30	8	Highest
29	44.2	0.137	77.9	4.01	4.78	4.41	4.40	5	Highest
30	37.2	0.471	13.1	1.99	1.22	4.41	2.54	104	Lowest
31	38.2	0.431	19.8	2.28	1.64	4.41	2.78	99	Lowest
32	41.5	0.394	48.3	3.23	2.05	4.27	3.18	83	Low
33	43.1	0.269	65.5	3.72	3.38	4.84	3.98	42	High
34	41.7	0.271	56.7	3.31	3.35	4.53	3.73	56	Medium
35	38.5	0.467	15.3	2.37	1.26	4.41	2.68	100	Lowest
36	43.5	0.250	70.1	3.81	3.58	4.57	3.99	41	High
37	44.3	0.221	82.1	4.05	3.89	4.14	4.03	36	High
38	43.9	0.198	95.4	3.94	4.13	4.41	4.16	20	Highest
39	38.5	0.410	37.5	2.38	1.88	4.55	2.94	93	Lowest
40	43.9	0.192	84.9	3.94	4.20	4.14	4.09	28	High
41	34.7	0.442	6.0	1.27	1.53	4.26	2.35	109	Lowest

Ward	LST (°C)	NDVI	Built-up (%)	Heat	Green def.	Flood	UCSI	Rank	Class
42	34.0	0.418	6.0	1.07	1.79	4.26	2.37	108	Lowest
43	44.8	0.189	95.8	4.21	4.23	4.14	4.19	16	Highest
44	45.4	0.173	97.4	4.35	4.40	3.87	4.21	14	Highest
45	43.9	0.220	89.6	3.94	3.89	4.14	3.99	40	High
46	43.9	0.216	86.2	3.94	3.94	4.14	4.01	37	High
47	44.9	0.184	86.5	4.21	4.28	3.87	4.12	24	High
48	46.1	0.202	78.8	4.56	4.09	3.56	4.07	30	High
49	43.7	0.249	72.0	3.87	3.59	4.14	3.87	45	High
50	45.0	0.197	90.5	4.25	4.14	4.14	4.18	19	Highest
51	42.1	0.254	76.0	3.42	3.54	4.41	3.79	49	Medium
52	34.9	0.430	7.4	1.34	1.66	4.26	2.42	106	Lowest
53	41.1	0.339	54.6	3.11	2.63	4.69	3.48	70	Low
54	42.2	0.327	71.5	3.44	2.76	4.41	3.54	66	Medium
55	44.6	0.221	72.5	4.13	3.89	4.25	4.09	29	High
56	44.2	0.270	56.3	4.03	3.37	4.69	4.03	34	High
57	45.8	0.176	84.1	4.49	4.36	4.57	4.48	2	Highest
58	41.9	0.303	82.9	3.36	3.01	4.41	3.60	65	Medium
59	38.8	0.373	27.9	2.45	2.27	4.39	3.04	90	Lowest
60	37.9	0.422	27.0	2.20	1.74	4.55	2.83	96	Lowest
61	41.6	0.266	82.3	3.28	3.41	4.14	3.61	64	Medium
62	43.0	0.266	83.3	3.66	3.40	4.14	3.74	55	Medium
63	42.4	0.257	82.6	3.50	3.51	4.14	3.72	58	Medium
64	38.1	0.263	33.6	2.26	3.44	4.55	3.42	71	Low
65	35.1	0.224	16.2	1.40	3.85	4.41	3.22	80	Low
66	45.3	0.170	86.8	4.34	4.43	4.41	4.39	6	Highest
67	46.4	0.181	80.8	4.67	4.31	4.30	4.43	3	Highest
68	45.9	0.158	74.6	4.50	4.56	3.29	4.12	25	High
69	43.8	0.243	22.6	3.91	3.65	2.26	3.27	79	Low
70	47.6	0.116	98.2	5.00	5.00	3.45	4.48	1	Highest
71	46.0	0.206	63.8	4.53	4.05	3.29	3.95	43	High
72	46.3	0.209	75.3	4.64	4.01	3.99	4.21	13	Highest
73	42.7	0.235	35.0	3.58	3.74	2.26	3.19	82	Low
74	44.1	0.225	62.7	3.99	3.84	4.14	3.99	39	High
75	45.6	0.186	86.7	4.41	4.25	4.14	4.27	11	Highest
76	43.4	0.303	66.8	3.78	3.01	4.57	3.79	50	Medium
77	42.3	0.238	73.2	3.47	3.71	4.84	4.00	38	High
78	40.4	0.385	28.6	2.91	2.14	4.55	3.20	81	Low
79	43.0	0.284	56.6	3.67	3.22	4.69	3.86	47	Medium
80	42.8	0.264	23.4	3.62	3.43	2.26	3.10	86	Low
81	44.7	0.192	91.0	4.17	4.20	3.72	4.03	35	High
82	43.0	0.301	53.1	3.68	3.03	4.27	3.66	61	Medium
83	42.9	0.270	75.7	3.64	3.36	4.41	3.81	48	Medium
84	35.9	0.410	12.6	1.62	1.87	4.41	2.63	102	Lowest
85	37.7	0.387	24.7	2.14	2.12	4.55	2.94	94	Lowest
86	44.2	0.190	91.6	4.03	4.22	4.14	4.13	23	Highest

Ward	LST (°C)	NDVI	Built-up (%)	Heat	Green def.	Flood	UCSI	Rank	Class
87	44.4	0.259	82.3	4.09	3.48	3.72	3.76	53	Medium
88	43.5	0.306	74.3	3.83	2.98	3.72	3.51	69	Low
89	45.7	0.125	95.2	4.46	4.91	3.56	4.31	7	Highest
90	42.8	0.307	71.8	3.61	2.97	3.56	3.38	73	Low
91	42.0	0.352	59.9	3.38	2.49	3.41	3.09	87	Low
92	42.4	0.312	70.9	3.50	2.92	3.56	3.33	77	Low
93	44.7	0.143	86.8	4.15	4.72	3.56	4.14	22	Highest
94	44.5	0.203	96.7	4.11	4.08	3.99	4.06	31	High
95	44.5	0.185	83.1	4.11	4.27	3.72	4.03	32	High
96	34.2	0.470	11.3	1.13	1.23	4.41	2.25	111	Lowest
97	36.4	0.431	21.4	1.76	1.65	4.55	2.65	101	Lowest
98	39.1	0.310	23.3	2.56	2.94	4.39	3.29	78	Low
99	43.1	0.311	54.8	3.71	2.93	4.69	3.78	52	Medium
100	43.1	0.292	49.4	3.69	3.13	4.11	3.64	63	Medium
101	42.8	0.340	37.2	3.62	2.62	2.68	2.97	92	Lowest
102	40.9	0.423	35.6	3.08	1.74	3.00	2.60	103	Lowest
103	40.3	0.410	32.5	2.90	1.87	2.10	2.29	110	Lowest
104	40.7	0.432	41.1	3.01	1.64	2.56	2.40	107	Lowest
105	39.6	0.176	50.5	2.70	4.36	4.11	3.72	57	Medium
106	43.8	0.212	69.2	3.92	3.99	4.41	4.11	27	High
107	37.2	0.228	14.9	2.00	3.81	4.41	3.41	72	Low
108	38.2	0.417	25.3	2.29	1.79	4.39	2.82	97	Lowest
109	42.6	0.237	81.0	3.56	3.72	4.41	3.90	44	High
110	38.7	0.341	13.1	2.44	2.61	4.41	3.15	85	Low
111	33.7	0.252	8.2	1.00	3.56	4.42	2.99	91	Lowest

Appendix B. Google Earth Engine scripts

The five scripts are in the folder gee/. Register free at code.earthengine.google.com (non-commercial use), upload the wards as an asset named navi_mumbai_wards, paste a script, press Run and start each export from the Tasks tab.

Script	Data set (Earth Engine ID)	Output
01_LST_Landsat_Apr2026.js	LANDSAT/LC08/C02/T1_L2 and LC09 (ST_B10, QA_PIXEL)	LST_Apr2026_C.tif (30 m)
02_NDVI_Sentinel2_Apr2026.js	COPERNICUS/S2_SR_HARMONIZED (B4, B8, SCL)	NDVI_Apr2026.tif (10 m)
03_BuiltUp_LandCover.js	GOOGLE/DYNAMICWORLD/V1 and ESA/WorldCover/v200	Built-up share, land-cover raster
04_Elevation_Slope_WaterDistance.js	USGS/SRTMGL1_003 and WorldCover water classes	Elevation, slope, distance to water
05_Ward_Zonal_Statistics.js	All of the above, reduced over the ward asset	Ward_stats_30m.csv, Ward_stats_10m.csv

Values to expect when the scripts are run: about 3 Landsat scenes (13, 22, 30 April 2026); mean LST inside the study area about 41 °C; about 2 Sentinel-2 scenes (8 and 11 April); mean NDVI about 0.30; built-up share about 33%; ward 70 the hottest ward (47 to 48 °C).

Script 01 (land surface temperature), as an example

```
// ============================================================================
// 01  Land Surface Temperature (LST), Navi Mumbai, April 2026
// Data: Landsat 8 + 9 Collection 2 Level 2, band ST_B10 (surface temperature)
// Output: LST_Apr2026_C.tif (30 m, EPSG:32643) in Google Drive folder NaviMumbai_GEE
// BEFORE RUNNING: upload the ward polygons as an Earth Engine asset (see README.txt)
// ============================================================================
var wards = ee.FeatureCollection('projects/YOUR_PROJECT/assets/navi_mumbai_wards'); // <-- change to your asset path
var region = wards.geometry().dissolve(50);
Map.centerObject(region, 11);

// Clear April 2026 scenes: 13, 22 and 30 April (cloud cover < 20%)
var start = '2026-04-10', end = '2026-05-01';   // end date is exclusive

function prepLandsat(img) {
  // QA_PIXEL bits: 1 = dilated cloud, 2 = cirrus, 3 = cloud, 4 = cloud shadow
  var qa = img.select('QA_PIXEL');
  var clear = qa.bitwiseAnd(1 << 1).eq(0)
      .and(qa.bitwiseAnd(1 << 2).eq(0))
      .and(qa.bitwiseAnd(1 << 3).eq(0))
      .and(qa.bitwiseAnd(1 << 4).eq(0));
  // Scale factors from the Earth Engine catalog: ST = DN * 0.00341802 + 149.0 (Kelvin)
  var lstC = img.select('ST_B10').multiply(0.00341802).add(149.0).subtract(273.15).rename('LST_C');
  return lstC.updateMask(clear).copyProperties(img, ['system:time_start']);
}

var l8 = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2');
var l9 = ee.ImageCollection('LANDSAT/LC09/C02/T1_L2');
var scenes = l8.merge(l9).filterBounds(region).filterDate(start, end)
    .filter(ee.Filter.lt('CLOUD_COVER', 20));
print('Landsat scenes used (expect about 3):', scenes.size());
print('Scene dates:', scenes.aggregate_array('DATE_ACQUIRED'));

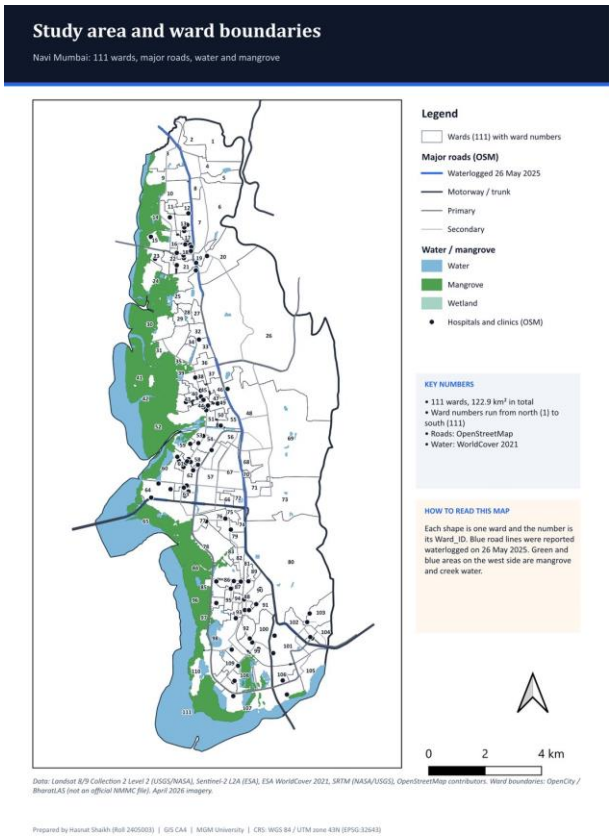
var lst = scenes.map(prepLandsat).median().clip(region);
lst = lst.updateMask(lst.gt(10).and(lst.lt(70)));   // remove impossible values

Map.addLayer(lst, {min: 32, max: 48, palette: ['#FFF4D6', '#FDD28A', '#F59E42', '#DC5A2B', '#8E1B1F']}, 'LST (C) April
2026');

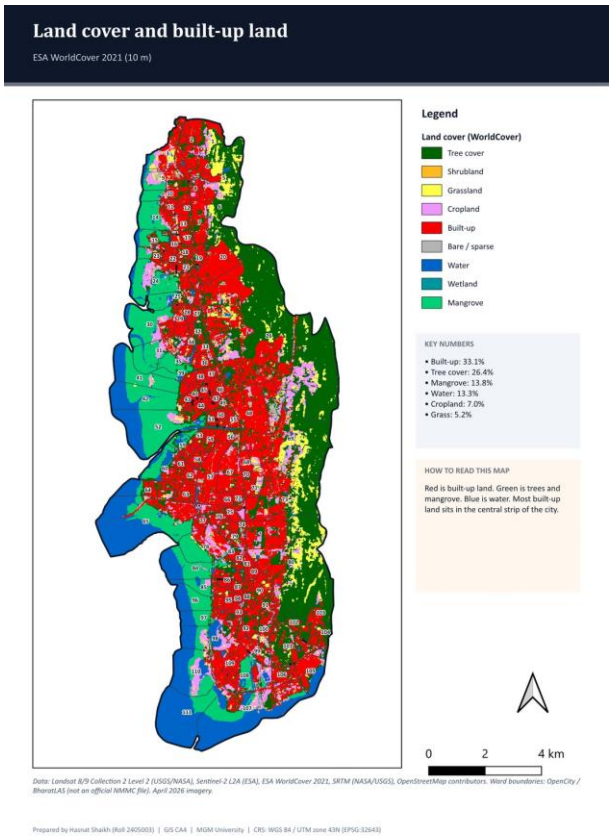
// Quick check: mean LST inside the study area (the delivered raster gives about 41.0 C)
print('Mean LST (C):', lst.reduceRegion({reducer: ee.Reducer.mean(), geometry: region, scale: 30, maxPixels: 1e9}));

Export.image.toDrive({image: lst, description: 'LST_Apr2026_C', folder: 'NaviMumbai_GEE',
  region: region, scale: 30, crs: 'EPSG:32643', maxPixels: 1e9});
```

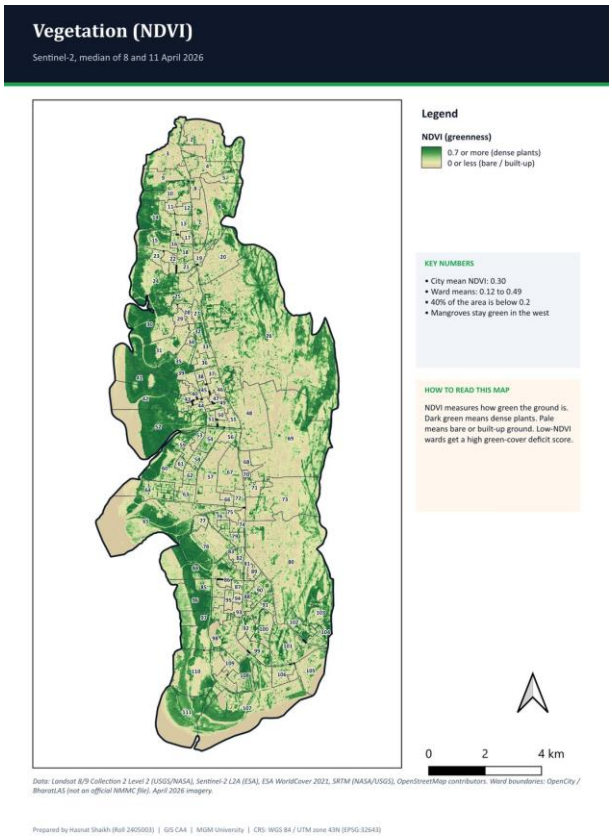
Appendix C. Thematic maps (1 to 10)



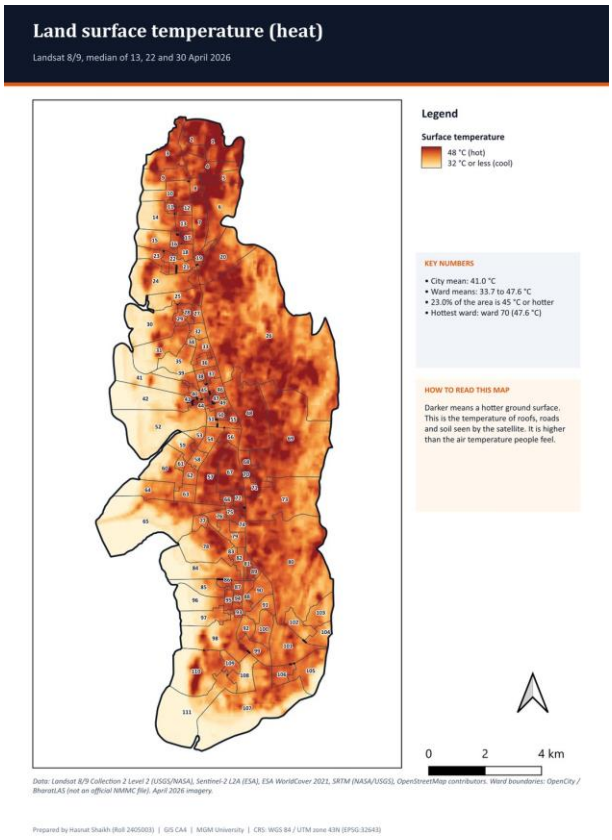
Map 1. Study area and wards



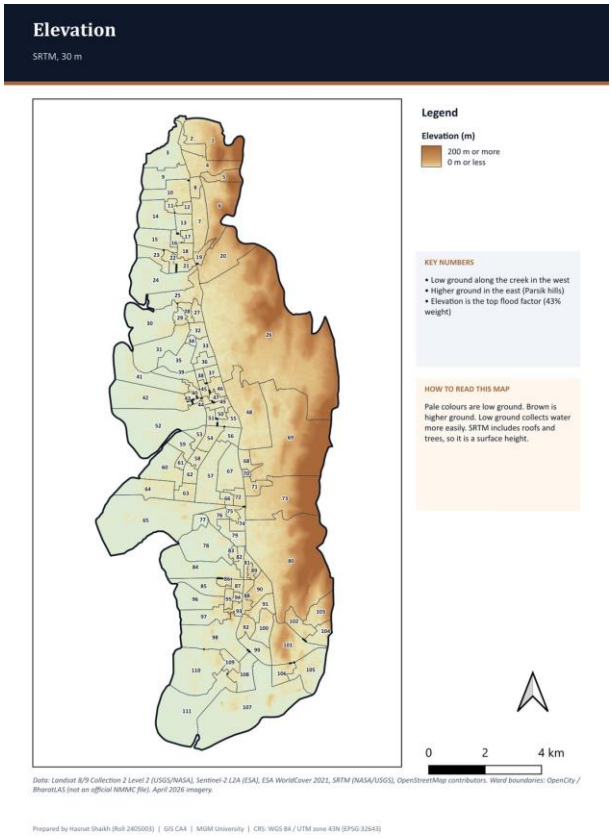
Map 2. Land cover



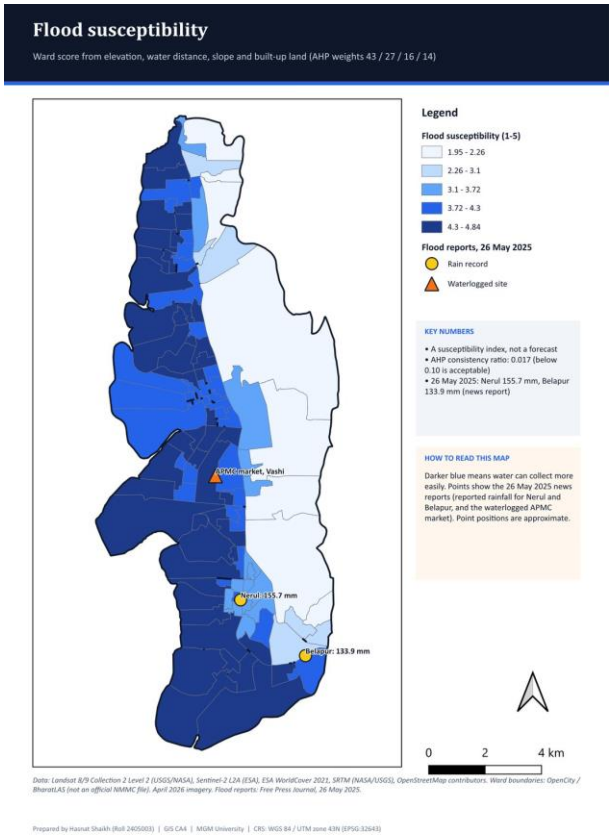
Map 3. Vegetation (NDVI)



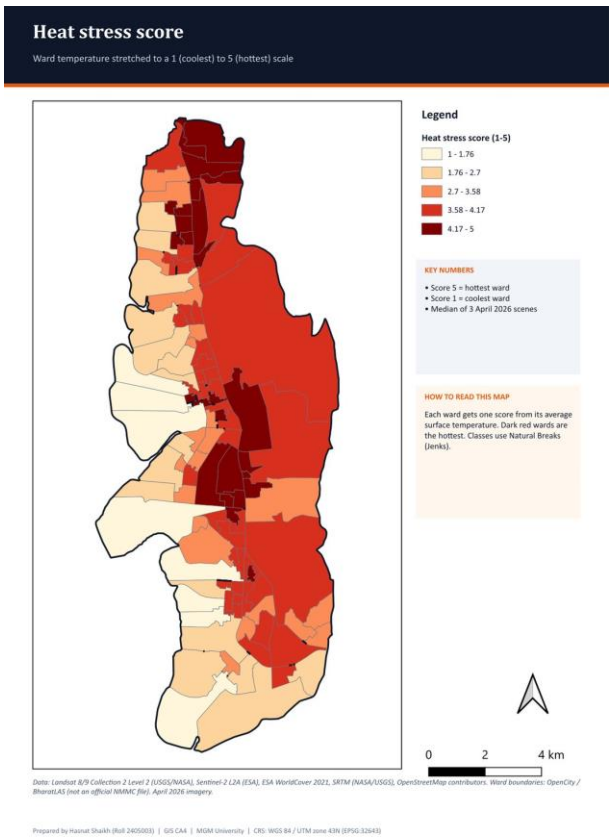
Map 4. Land surface temperature



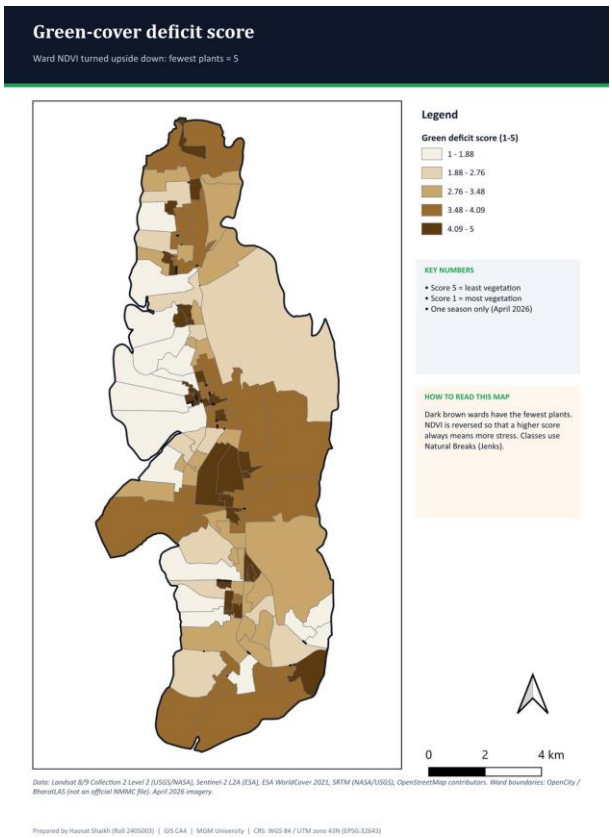
Map 5. Elevation



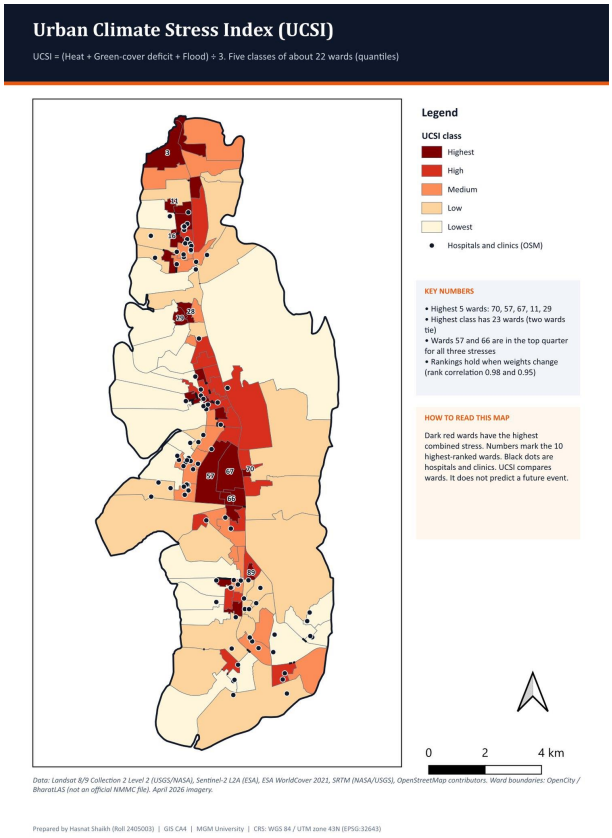
Map 6. Flood susceptibility



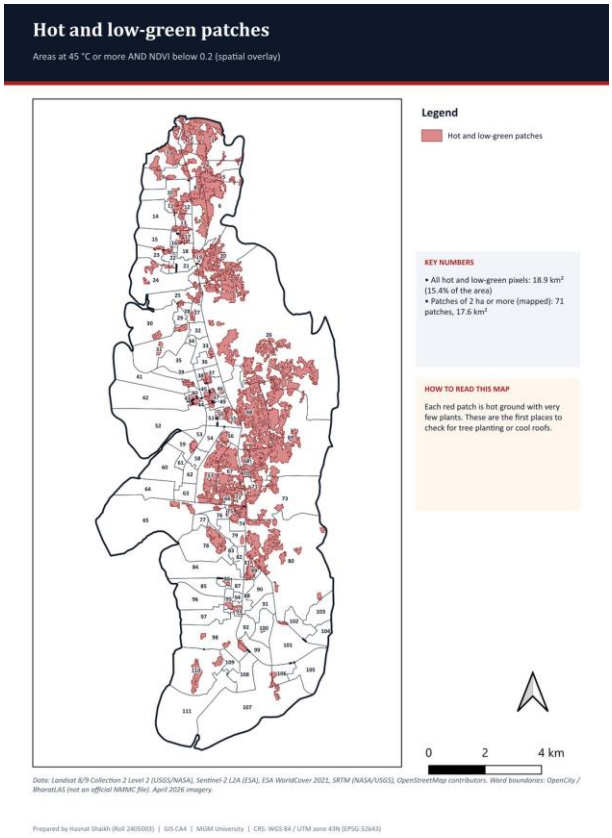
Map 7. Heat stress score



Map 8. Green-cover deficit score



Map 9. UCSI (final map)

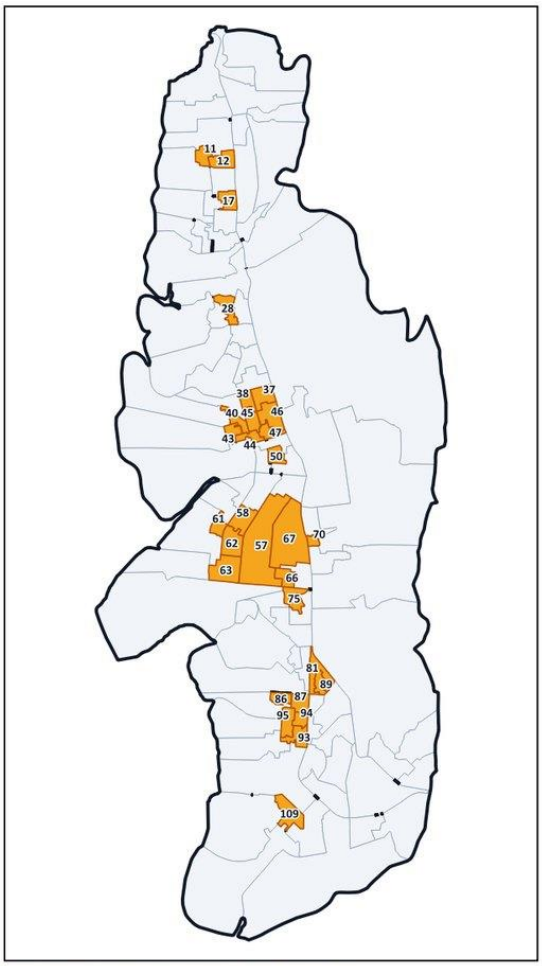


Map 10. Hot and low-green patches

Appendix D. Query results (screenshots)

Query A1: Built-up land 80% or more

30 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"BuiltUp_Pct" >= 80

QGIS tool: Processing > Extract by expression
```

RESULT

30 of 111 wards match

WHAT IT MEANS

30 wards are 80% or more built-up. Ward 70 is the most built-up (98.25%).

Result of the query

Other wards

0 2 4 km

Table: first 12 of 30 rows, sorted by Built-up %, highest first.

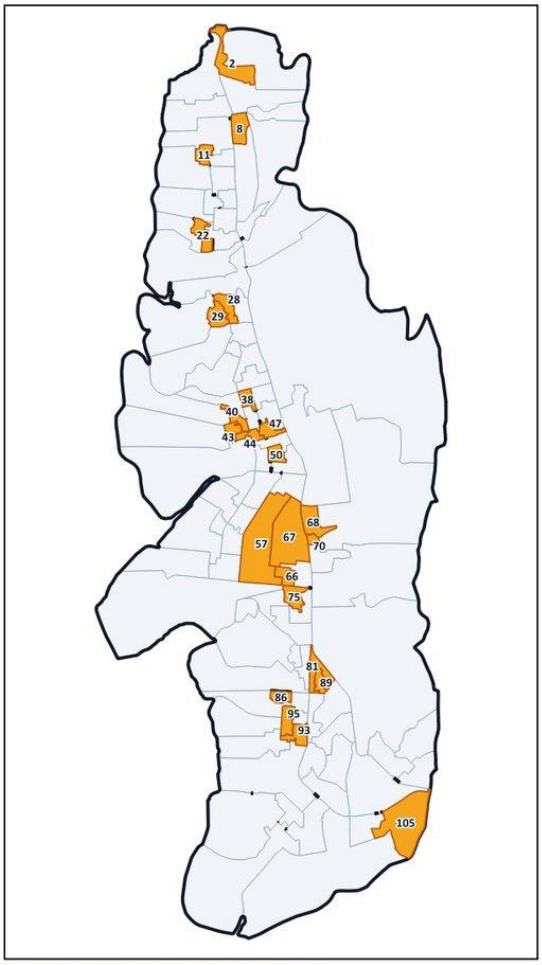
Ward_ID	BuiltUp_Pct	LST_C	UCSI
70	98.25	47.597	4.48
44	97.42	45.351	4.21
94	96.66	44.526	4.06
43	95.84	44.847	4.19
38	95.4	43.937	4.16
89	95.2	45.716	4.31
86	91.57	44.235	4.13
81	90.98	44.709	4.03
50	90.53	45.005	4.18
45	89.64	43.914	3.99
11	87.68	45.305	4.41
12	86.91	45.074	4.19

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query A1 Built-up 80% or more

Query A2: Vegetation index (NDVI) below 0.2

24 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"NDVI" < 0.2

QGIS tool: Processing > Extract by expression
```

RESULT

24 of 111 wards match

WHAT IT MEANS

24 wards have NDVI below 0.2, which means very few plants. Ward 70 has the lowest value (0.1164).

Result of the query

Other wards

0 2 4 km

Table: first 12 of 24 rows, sorted by NDVI, lowest first.

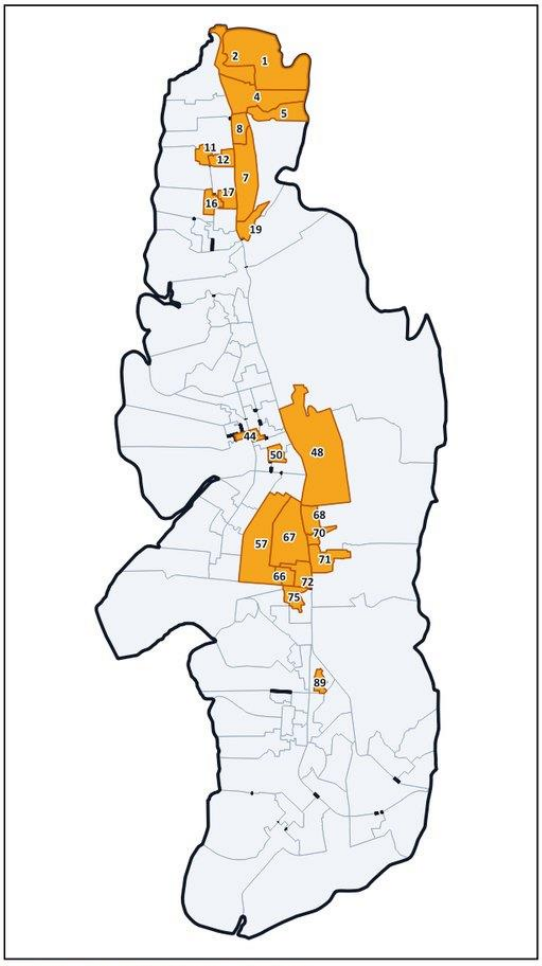
Ward_ID	NDVI	GreenCover_Pct	UCSI
70	0.1164	1.6	4.48
89	0.1252	3.4	4.31
29	0.1369	11.9	4.4
11	0.1399	9.5	4.41
93	0.1429	6.2	4.14
68	0.1575	11.8	4.12
2	0.1664	23.7	4.15
66	0.17	6.9	4.39
44	0.1729	2.2	4.21
57	0.1761	8.4	4.48
105	0.1763	15.8	3.72
28	0.1769	11.5	4.3

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query A2 NDVI below 0.2

Query A3: Surface temperature 45 °C or more

23 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"LST_C" >= 45

QGIS tool: Processing > Extract by expression
```

RESULT

23 of 111 wards match

WHAT IT MEANS

23 wards have an average ground surface temperature of 45 °C or more. Ward 70 is the hottest (47.6 °C). This is ground temperature, not air temperature.

Result of the query

Other wards

0 2 4 km

Table: first 12 of 23 rows, sorted by LST, hottest first.

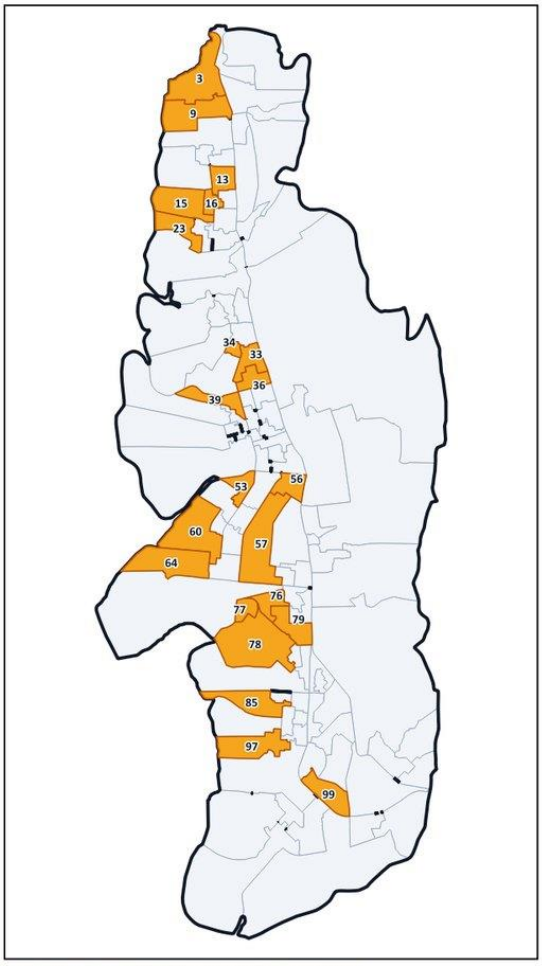
Ward_ID	LST_C	NDVI	BuiltUp_Pct
70	47.597	0.1164	98.25
1	46.972	0.2418	31.73
67	46.442	0.1807	80.85
72	46.346	0.209	75.3
48	46.057	0.2018	78.83
5	45.983	0.261	30.45
71	45.959	0.2056	63.83
68	45.88	0.1575	74.6
57	45.847	0.1761	84.13
8	45.733	0.1838	70.96
89	45.716	0.1252	95.2
4	45.692	0.2424	46.3

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query A3 LST 45 °C or more

Query A4: Flood susceptibility score 4.5 or more

22 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"Flood_Score" >= 4.5

QGIS tool: Processing > Extract by expression
```

RESULT

22 of 111 wards match

WHAT IT MEANS

22 wards score 4.5 or more out of 5. Susceptibility means water can collect there more easily. It is not a flood forecast.

Result of the query

Other wards

0 2 4 km

Table: first 12 of 22 rows, sorted by flood score, highest first.

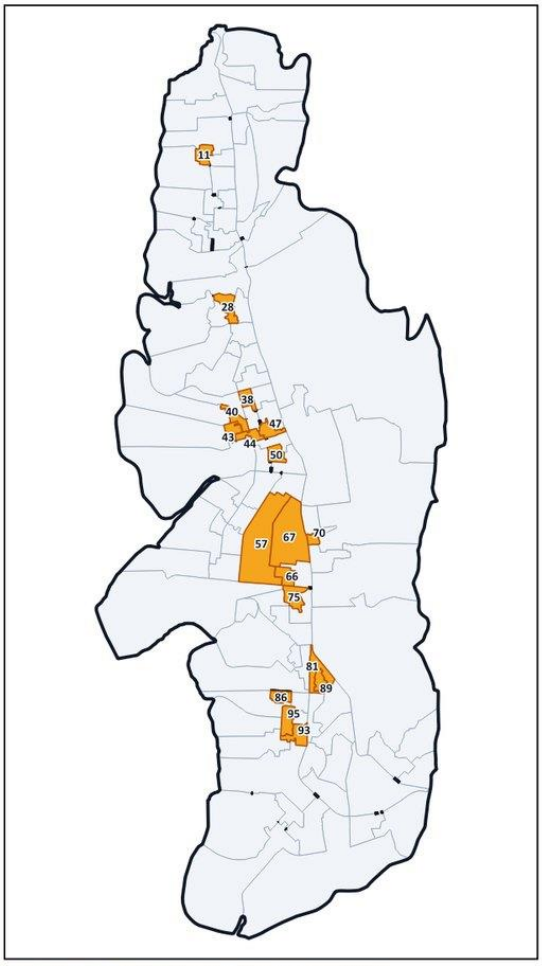
Ward_ID	Flood_Score	Elev_m	DistWater_m
16	4.84	4.94	143
33	4.84	4.04	191.5
77	4.84	1.87	147.9
3	4.69	2.79	158.2
53	4.69	4.24	110.6
56	4.69	4.57	166
79	4.69	4.1	165.8
99	4.69	4.27	222.6
13	4.57	4.95	325.4
36	4.57	4.12	303.3
57	4.57	4.88	285.6
76	4.57	4.68	408

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query A4 Flood score 4.5 or more

Query A5: Built-up 80% or more AND NDVI below 0.2

18 of 111 wards match



HOW IT WAS RUN IN QGIS

```
Select by Expression on layer 'wards':
"BuiltUp_Pct" >= 80 AND "NDVI" < 0.2

QGIS tool: Processing > Extract by expression
```

RESULT

18 of 111 wards match

WHAT IT MEANS

18 wards are both very built-up and very bare of plants. Ward 70 ranks highest in this group (UCSI rank 1 of 111).

Result of the query

Other wards

0 2 4 km

Table: first 12 of 18 rows, sorted by UCSI rank (1 = highest stress).

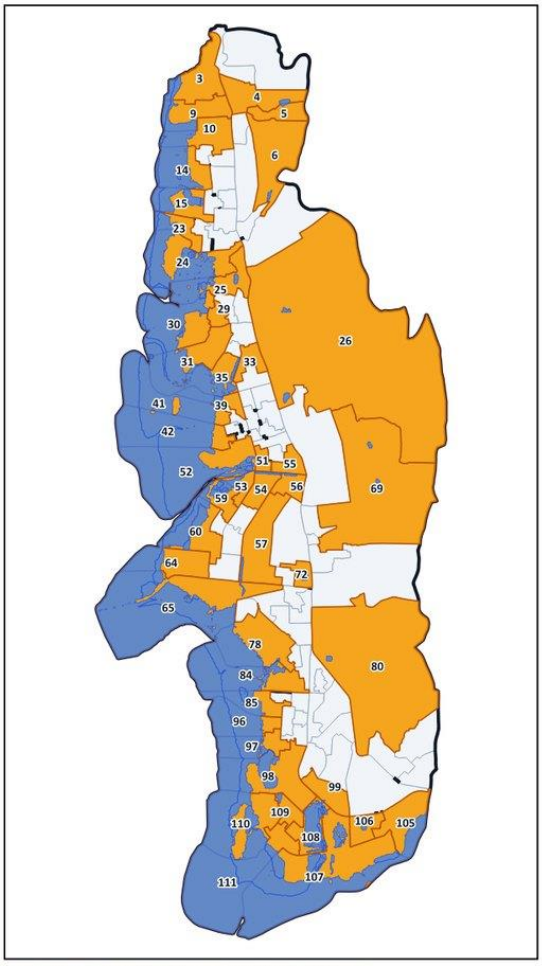
Ward_ID	BuiltUp_Pct	NDVI	LST_C	UCSI	UCSI_Rank
70	98.25	0.1164	47.597	4.48	1
57	84.13	0.1761	45.847	4.48	2
67	80.85	0.1807	46.442	4.43	3
11	87.68	0.1399	45.305	4.41	4
66	86.84	0.17	45.302	4.39	6
89	95.2	0.1252	45.716	4.31	7
28	84.01	0.1769	44.592	4.3	8
75	86.71	0.1864	45.565	4.27	11
44	97.42	0.1729	45.351	4.21	14
43	95.84	0.1888	44.847	4.19	16
50	90.53	0.1968	45.005	4.18	19
38	95.4	0.1982	43.937	4.16	20

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query A5 Built-up AND low NDVI

Query S1: Wards intersecting water or mangrove (1 ha or more)

48 of 111 wards



HOW IT WAS RUN IN QGIS

```
Vector > Research Tools > Select by Location
Input: wards
Predicate: intersect
Comparison: water_bodies where Area_ha >= 1
```

RESULT

48 of 111 wards

WHAT IT MEANS

48 wards touch a water or mangrove body of 1 ha or more (blue). Distance to water is one of the four flood factors, so these wards are the first to check.

Result of the query

Other wards

0 2 4 km

Table: first 12 of 48 rows, sorted by water share, highest first.

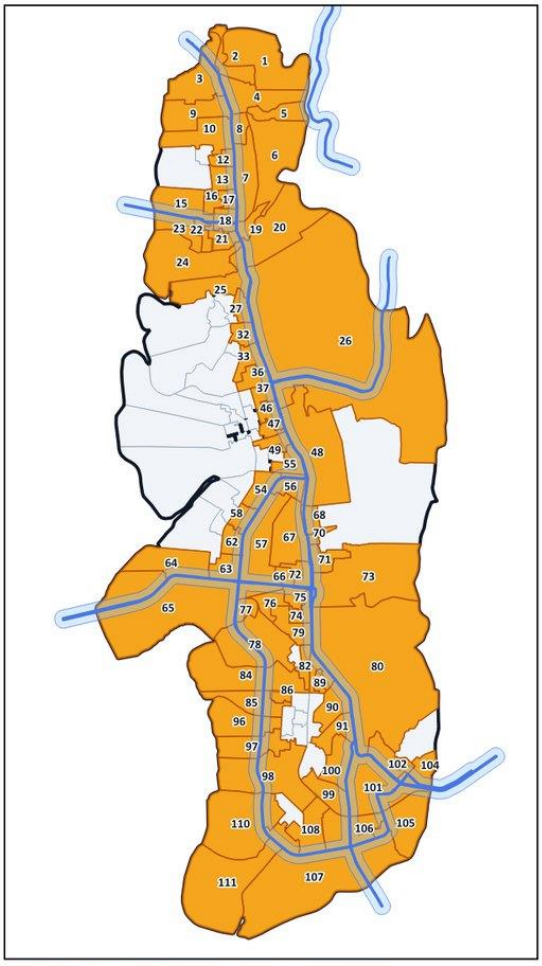
Ward_ID	Water_Pct	Flood_Score	UCSI
42	88.5	4.26	2.37
41	88.3	4.26	2.35
111	88.2	4.42	2.99
96	84.2	4.41	2.25
52	79.8	4.26	2.42
65	75.4	4.41	3.22
84	74.5	4.41	2.63
30	70.7	4.41	2.54
14	67.9	4.41	2.53
97	67.2	4.55	2.65
110	64.2	4.41	3.15
107	62.8	4.41	3.41

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query S1 Wards touching water

Query S2: Wards within 200 m of a major road

78 of 111 wards



HOW IT WAS RUN IN QGIS

```
Vector > Geoprocessing > Buffer
Input: major_roads (motorway, trunk, primary)
Distance: 200 m, dissolve
then Select by Location:
wards intersect the buffer
```

RESULT

78 of 111 wards

WHAT IT MEANS

78 wards have a motorway, trunk or primary road within 200 m (blue buffer). Two of these roads, Thane-Belapur Road and Sion-Panvel Highway, were reported waterlogged on 26 May 2025 (news report).

Table: first 12 of 78 rows, sorted by UCSI rank (1 = highest stress).

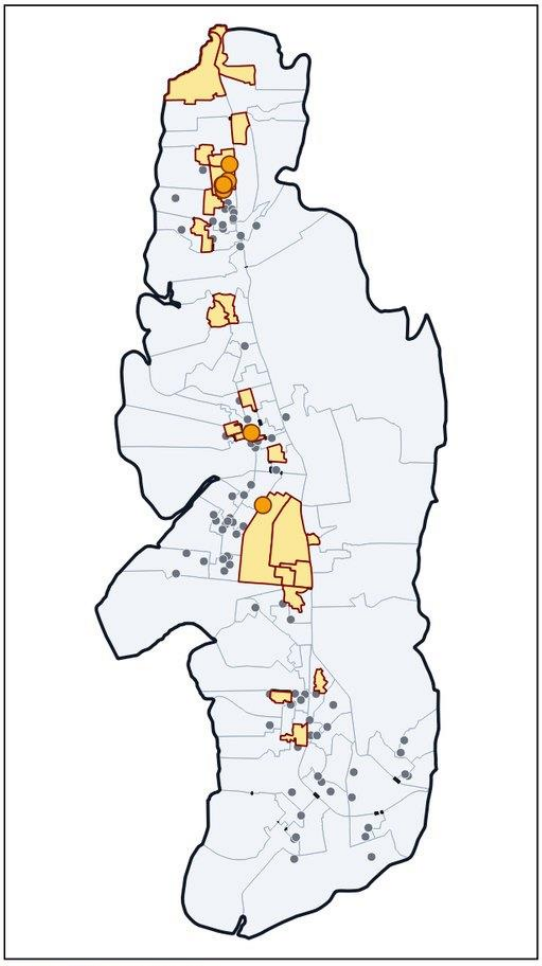
Ward_ID	BuiltUp_Pct	LST_C	UCSI	UCSI_Rank
70	98.25	47.597	4.48	1
57	84.13	45.847	4.48	2
67	80.85	46.442	4.43	3
66	86.84	45.302	4.39	6
89	95.2	45.716	4.31	7
3	47.19	44.653	4.29	9
16	69.18	45.07	4.27	10
75	86.71	45.565	4.27	11
8	70.96	45.733	4.24	12
72	75.3	46.346	4.21	13
22	78.05	44.055	4.2	15
12	86.91	45.074	4.19	17

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query S2 Wards within 200 m of roads

Query S3: Hospitals inside the highest-stress wards

8 of 93 mapped hospitals



HOW IT WAS RUN IN QGIS

```
Step 1: Select by expression
  "UCSI_Class" = 'Highest'  (23 wards)
Step 2: Select by Location
  Input: hospitals
  Predicate: are within
  Comparison: the 23 wards
```

RESULT

8 of 93 mapped hospitals

WHAT IT MEANS

Only 8 of the 93 mapped hospitals and clinics are inside the 23 highest-stress wards (yellow). OpenStreetMap may be missing some facilities, so treat this as a first check.

Result of the query

Other wards

0 2 4 km

Table: all 8 rows, sorted by Ward_ID.

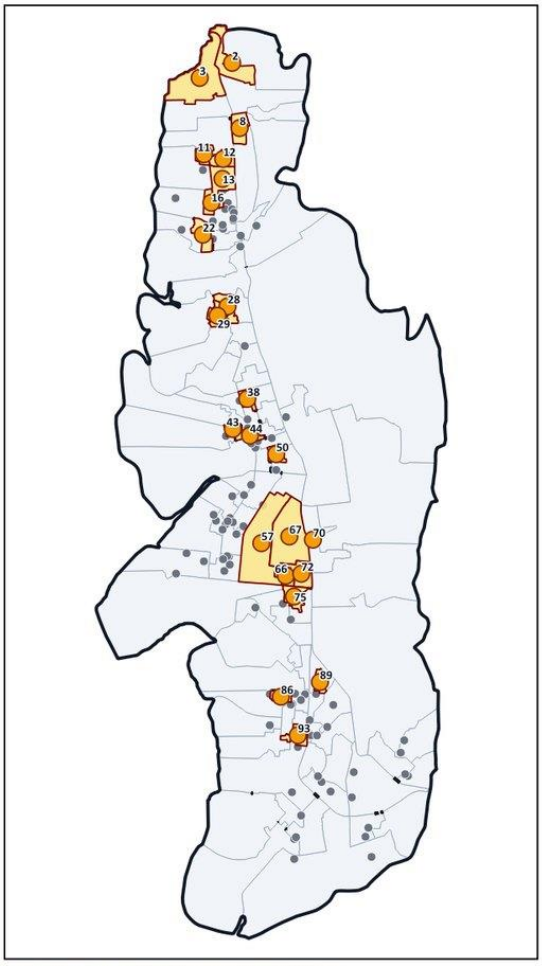
Hosp_ID	Name	Type	Ward_ID
38	Mody Hospital Iccu	hospital	12
6	Sai Seva Hospital and Icu, Mumbai	hospital	13
15	Indravati Hospital	hospital	13
50	Sanjivni Hospital, Thane	hospital	13
61	NMMC MCH Airoli	hospital	13
91	Unnamed hospital	hospital	13
44	Aakashdeep Hospital and I.C.U	hospital	44
71	Healing Hands Clinic - Vashi, New Mumbai	hospital	57

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query S3 Hospitals in highest wards

Query S4: Nearest hospital to each highest-stress ward

mean 693 m, longest 2505 m



HOW IT WAS RUN IN QGIS

```
Vector > Geoprocessing > Centroids
  (the highest-stress wards)
Vector > Data Management >
  Join attributes by nearest
  Input: ward centroids
  Join layer: hospitals, 1 neighbour
Field Dist_m = round(distance), metres
```

RESULT

mean 693 m, longest 2505 m

WHAT IT MEANS

From the centre of each highest-stress ward, the nearest mapped hospital is 693 m away on average. The longest distance is 2505 m (ward 2). Distances are straight lines, not road distance.

Result of the query

Other wards

0 2 4 km

Table: first 12 of 23 rows, sorted by distance, longest first.

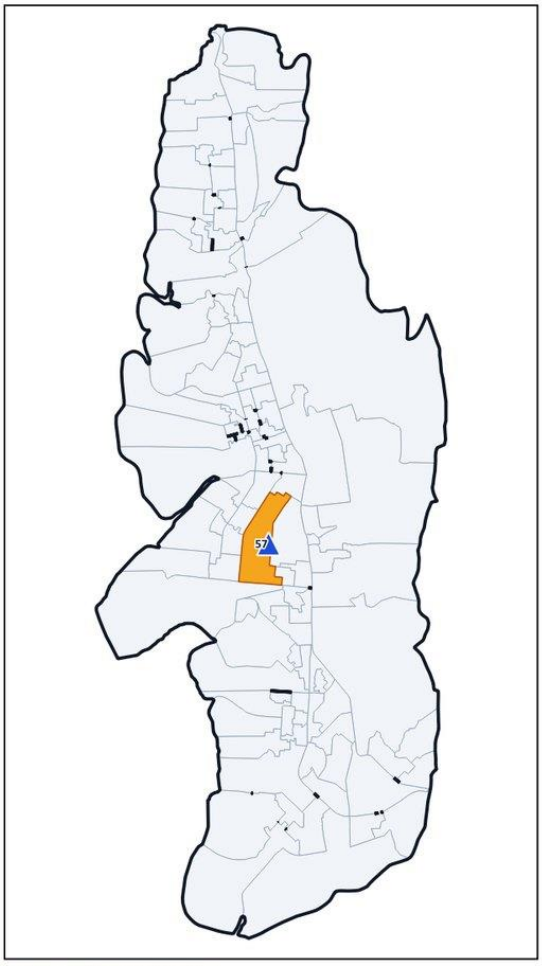
Ward_ID	Hosp_Name	Dist_m	UCSI
2	Mody Hospital Iccu	2,505	4.15
3	Mody Hospital Iccu	2,257	4.29
70	Healing Hands Clinic - Vashi, New Mumbai	1,496	4.48
28	Divine Multispecialty Hospital and Research Centre	1,067	4.3
67	Healing Hands Clinic - Vashi, New Mumbai	1,009	4.43
29	Divine Multispecialty Hospital and Research Centre	1,003	4.4
8	Mody Hospital Iccu	930	4.24
72	Sanpada General Hopital	868	4.21
66	Sanpada General Hopital	694	4.39
57	Pkc Hospital	608	4.48
11	Avadhoot Hospital and Iccu	392	4.41
50	Unnamed hospital	390	4.18

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query S4 Nearest hospital

Query S5: Ward that contains the waterlogged APMC market

Ward 57: flood score 4.57, UCSI 4.48 (Highest)



HOW IT WAS RUN IN QGIS

```
Vector > Research Tools > Select by Location
Input: wards
Predicate: contain
Comparison: flood_reports where
  Report_Type = 'Waterlogged site'
```

RESULT

Ward 57: flood score 4.57, UCSI 4.48 (Highest)

WHAT IT MEANS

The APMC market site reported waterlogged on 26 May 2025 lies in ward 57. The model gives this ward a high flood score (4.57) and UCSI rank 2 of 111, so the news report and the model agree here. This is a small consistency check, not proof. The point is approximate (centre of the market yards in OpenStreetMap).

Result of the query

Other wards

0 2 4 km

Table: all 1 rows, sorted by Ward_ID.

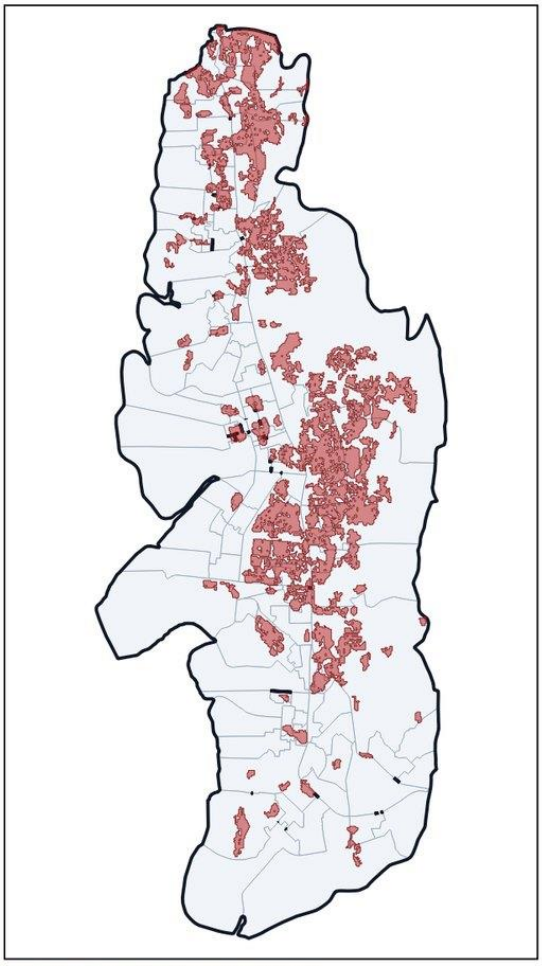
Ward_ID	Flood_Score	UCSI	UCSI_Class	UCSI_Rank
57	4.57	4.48	Highest	2

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query S5 Ward with waterlogged site

Query S6: Where hot and low-green land overlaps, ward by ward

17.6 km² in 71 patches; 85 wards touched



HOW IT WAS RUN IN QGIS

```
Raster calculator:
  LST >= 45 AND NDVI < 0.2
Polygonise, keep patches >= 2 ha
Vector > Geoprocessing > Intersection:
  wards with hot patches
Field calculator: $area / 10000 (ha)
Statistics by category: sum per ward
```

RESULT

17.6 km² in 71 patches; 85 wards touched

WHAT IT MEANS

Ward 26 has the largest hot and low-green area (281 ha). Among wards larger than 50 ha, ward 67 has the highest share (60.3% of the ward). These are the first places to look at for trees or cool roofs.

Hot and low-green patch (2 ha or more)

Other wards

0 2 4 km

Table: first 12 of 85 rows, sorted by area, largest first.

Ward_ID	Hot_lowgreen_ha	Ward_area_ha	Share_pct
26	281.5	1,686.2	16.7
69	191	647.1	29.5
48	147.1	290.8	50.6
80	107.3	802	13.4
1	82	201.7	40.6
57	80.2	155.5	51.5
67	71.2	118.1	60.3
20	60.8	194.5	31.2
6	56.5	195.1	29
3	44.1	123.4	35.8
7	41.5	99.7	41.7
73	35.1	326.7	10.7

Prepared by Hasnat Shaikh (Roll 2405003) | GIS CA4 | Result rendered by QGIS from NaviMumbai_UCSI.gpkg and saved in NaviMumbai_query_results.gpkg

Query S6 Hot and low-green overlay